NAME: TANKISO MASOEBE

INTRO T0 DATABASES WK 9

1.SCENARIO ANALYSIS:

A company uses a **Customer Relationship Management (CRM)** system to store information about customers and their orders. The CRM database contains two tables: **Customers** and **Orders**. The Customers table stores details such as CustomerID, Name, Email, and PhoneNumber. The Orders table stores OrderID, CustomerID, ProductName, OrderDate, and Amount. A sales manager wants to see a report that shows which customers placed orders and what products they purchased. To generate this report, the system uses a **JOIN query** to combine data from both tables using the common field **CustomerID**. By performing an **INNER JOIN**, the query links each order with the corresponding customer information. This allows the report to display the customer's name together with the product they ordered and the order date. Using a join makes the data more meaningful because it connects related information from multiple tables, helping the company analyze customer buying patterns and improve sales strategies.

2.CONCEPT RESEARCH:

An **Inner Join** is a database operation used in SQL to combine data from two or more tables based on a related column between them. It returns only the records that have matching values in both tables. Inner joins are commonly used when information is stored in separate tables but needs to be viewed together. For example, a **Customers** table may store customer details, while an **Orders** table stores purchase information. By using an inner join with a shared field such as **CustomerID**, the query links the tables and displays only customers who have placed orders. This helps users retrieve accurate and meaningful combined data.

3.TOOL PRACTICE:

Running a **LEFT JOIN** in **MySQL** helps combine data from two tables while keeping all records from the left table. During practice, I created two tables called **Customers** and **Orders** and ran a LEFT JOIN using the **CustomerID** field. The query displayed all customers from the Customers table, even those who had not placed any orders, while matching orders appeared where available. If a customer had no order, the order fields showed **NULL** values. This practice helped me understand how LEFT JOIN is useful for identifying missing relationships in data, such as customers who have not yet made purchases.

4.APPLICATION PRACTICE:

In **MySQL**, **INNER JOIN** and **LEFT JOIN** are used to combine data from related tables.

INNER JOIN query:

```
SELECT Customers.Name, Orders.ProductName
FROM Customers
INNER JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

LEFT JOIN query:

```
SELECT Customers.Name, Orders.ProductName
FROM Customers
LEFT JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

The **INNER JOIN** returns only records where a match exists in both tables, meaning it shows customers who placed orders. The **LEFT JOIN** returns all customers from the Customers table and any matching orders. If a customer has no order, the order fields appear as **NULL**, helping identify customers without purchases.